Computer Networks

Lecture # 05

Recap of Lecture 04

- Categories of Networks
  - PAN
  - LAN
  - MAN
  - WAN
  - Interconnection of Networks: Internetworks
  - Internet

Overview of Lecture 05

- The OSI Model
- Layered task
- Peer-to-Peer communication
- OSI Layers

THE OSI MODEL

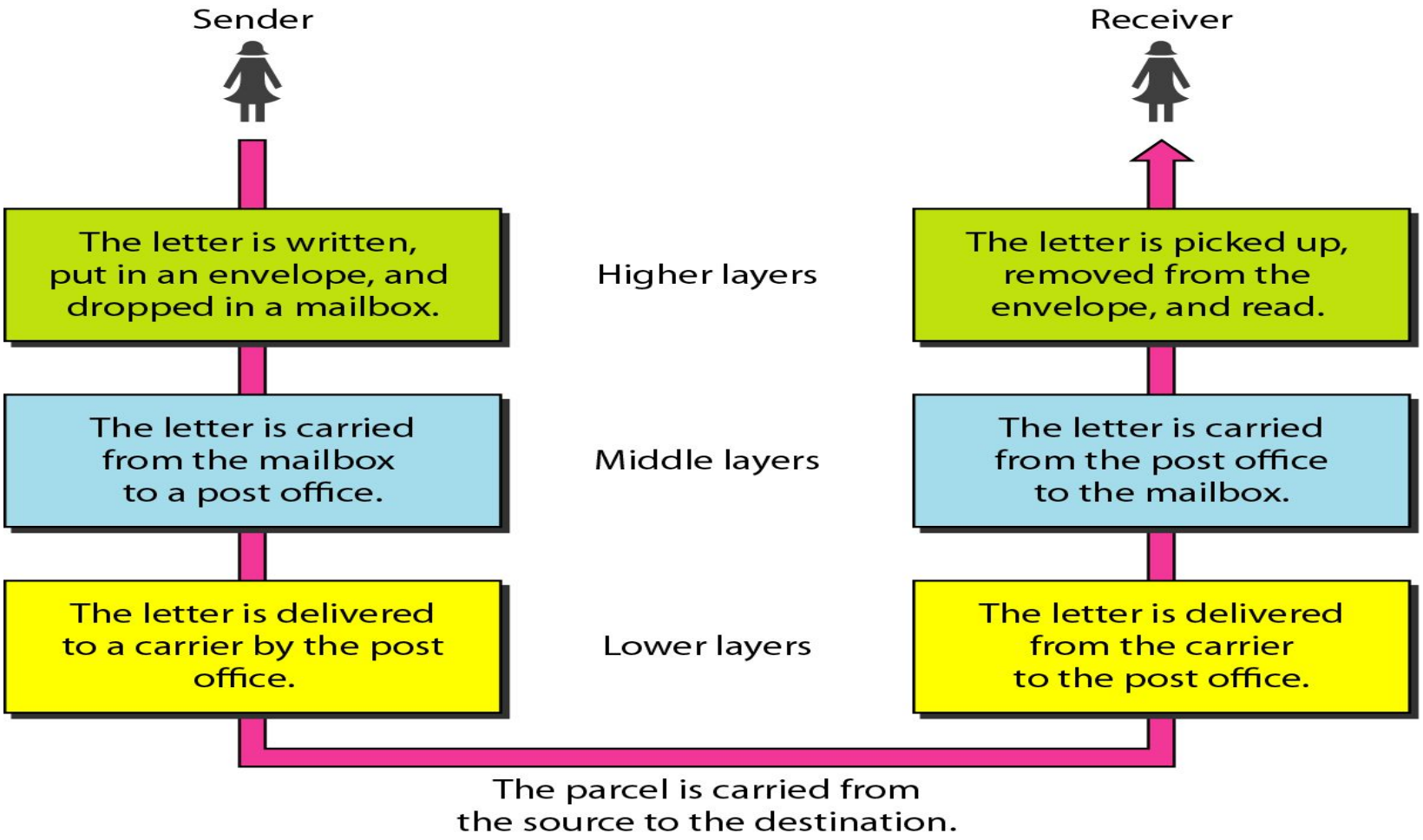
- The layered model that dominated data communications and networking literature before 1990 was the Open Systems Interconnection (OSI) model.
- Everyone believed that the OSI model would become the ultimate standard for data communications, but this did not happen.
- The TCPIIP protocol suite became the dominant commercial architecture because it was used and tested extensively in the Internet; the OSI model was never fully implemented.
- Established in 1947 by International Standards Organization (ISO)
- An ISO standard, that covers all aspects of network communications is the Open Systems Interconnection (OSI) model.
- It was first introduced in the late 1970s.
- No physical existence, but it is conceptual that helps to understand the complex methods of data transmission among various devices.

THE OSI MODEL Help in Understanding

- How physically computers communicate and connect.
- How bits are presented in a network medium.
- How this ensure that data will reach on the target error free and it will be received by correct computer.
- How network devices communicate, and how data is converted from one format to another.
- How computer comes to know that when it should transmit the data and when not.
- How network devices maintain proper rate of data flow.

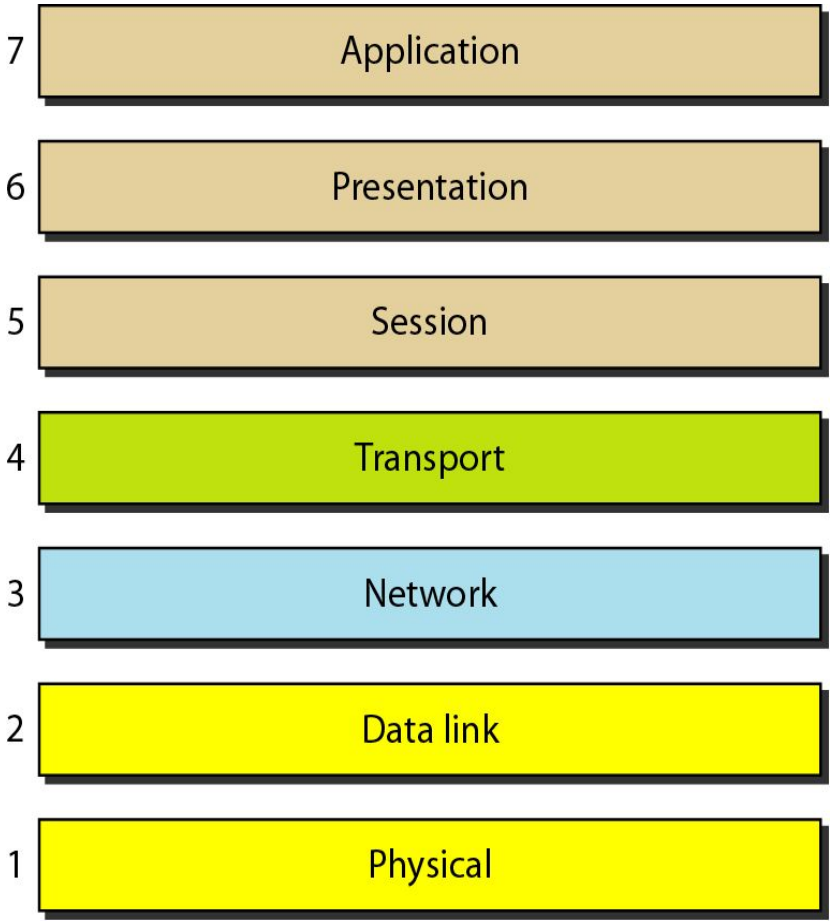
Layered Tasks

- Sender, Receiver and Carrier



Layered Architecture

Seven layers of the OSI model



Why layers ?

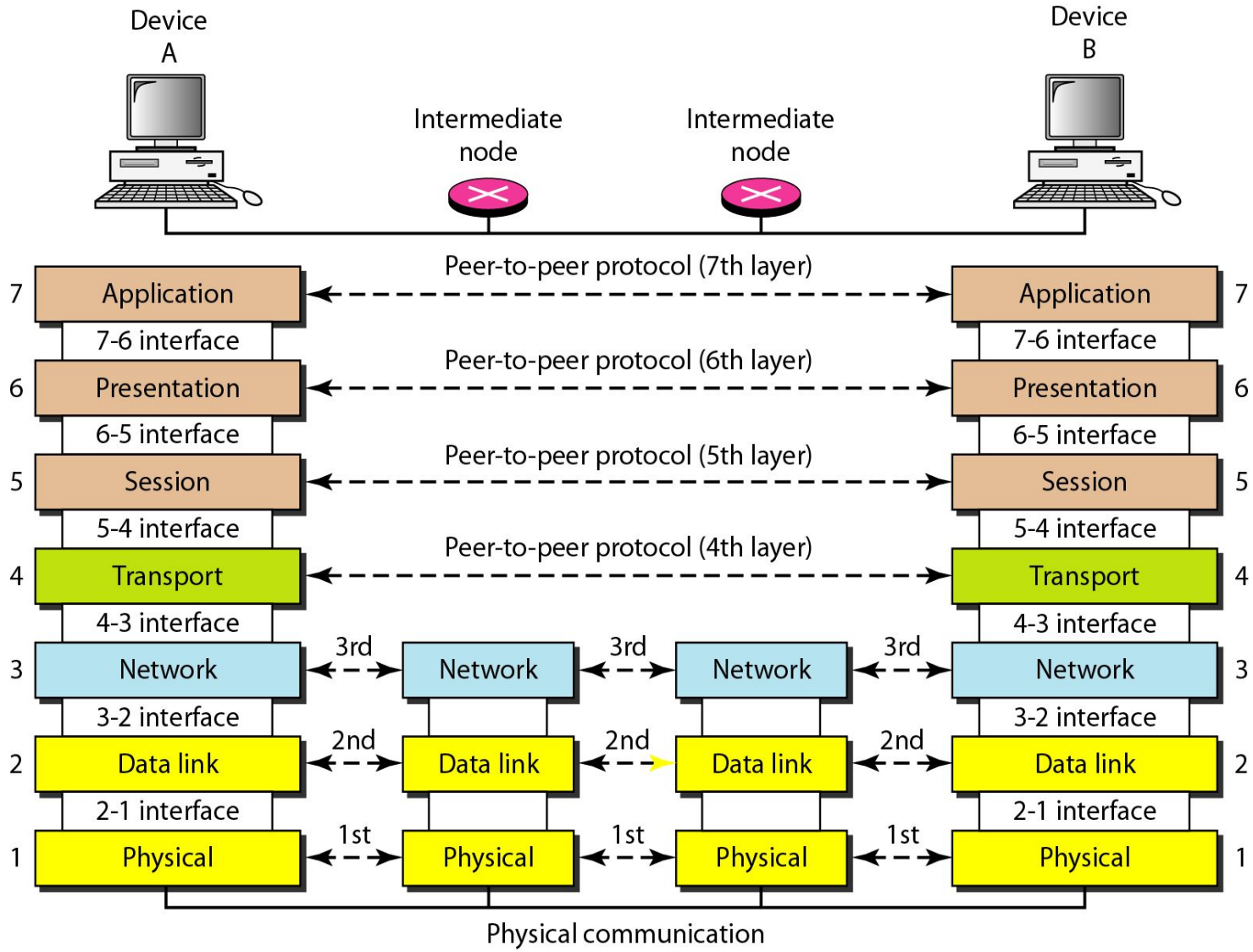
- Designer identified which networking functions had related uses and collected those functions into discrete groups that became the layers.
- The OSI model allows complete interoperability between otherwise incompatible systems.
- Each layer uses the services of the layer immediately below it and provides services to its above layer.

Peer-to-peer Processes

- Layer x on one machine communicates with layer x on another machine - called Peer-to-Peer Processes.
- Interfaces between Layers
  - Each interface defines what information and services a layer must provide for the layer above it.
  - Well defined interfaces and layer functions provide modularity to a network

Peer-to-peer Processes (cont'd)

The interaction between layers in the OSI model



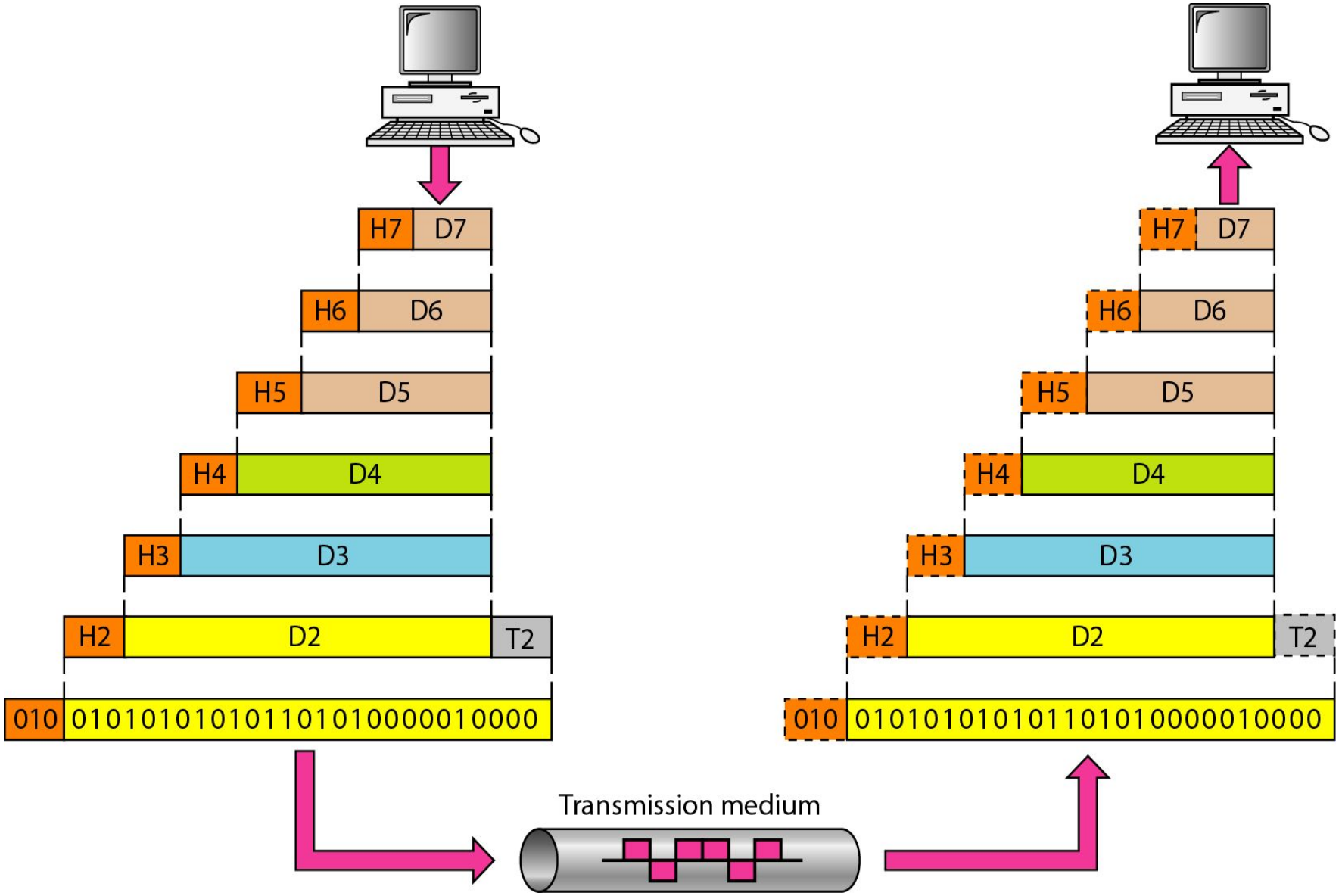
Organizations of the layers

1. Network support layers : Layers 1, 2, 3

2. User support layer : Layer 5, 6, 7
   - It allows interoperability among unrelated software systems

3. Transport layer (Layer 4) : links the two subgroups and ensures that what the lower layers have transmitted is in a form that the upper layers can use.

An exchange using the OSI model



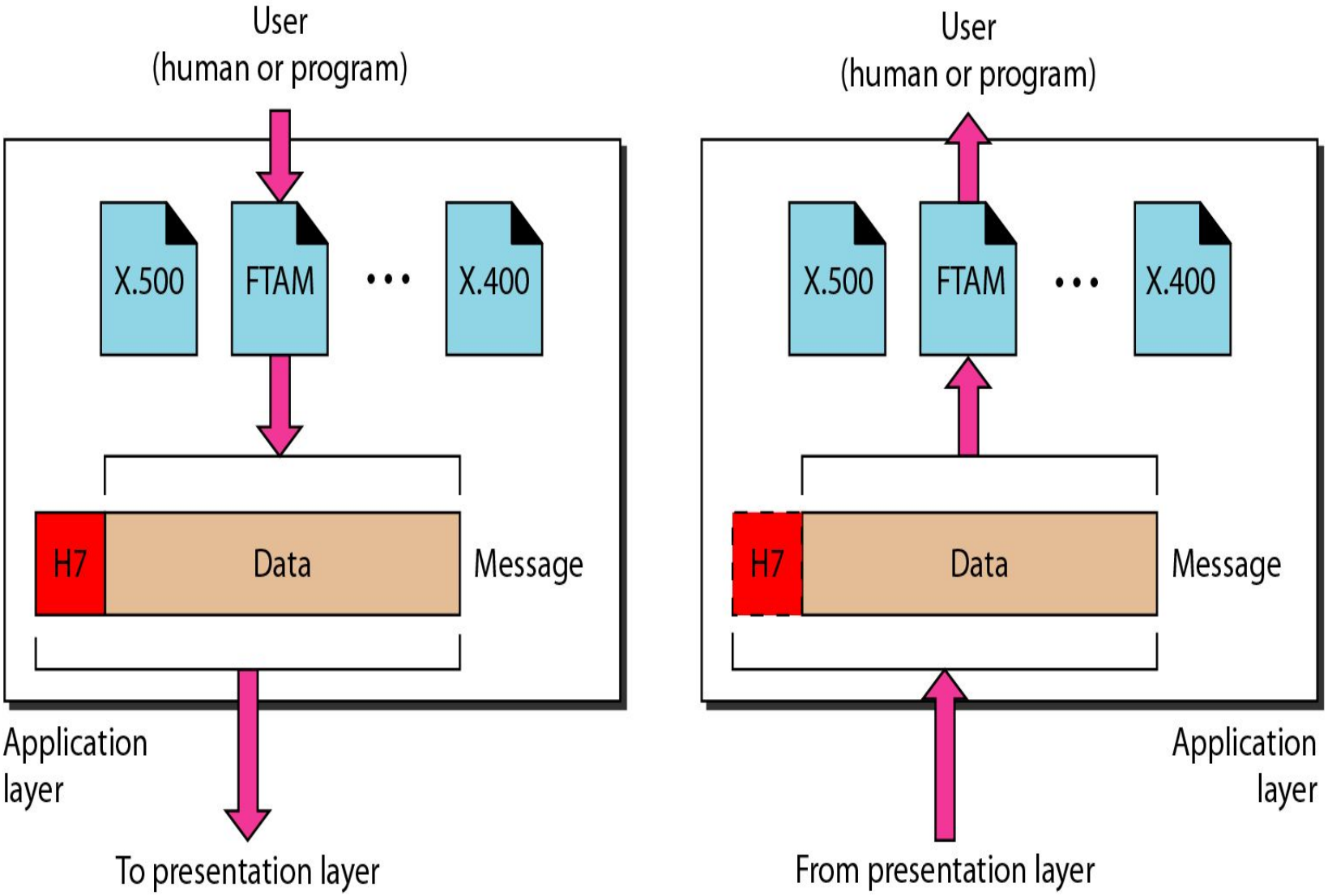
OSI Layers

Application Layer

- It enables the user, whether human or software, to access the network.

- It provides user interfaces and support for services such as
  - Electronic mail,
  - File Transfer, Access and management (FTAM),
  - Shared database management,
  - Network virtual terminal
  - Directory services (X.500)
  - and other types of distributed information services.

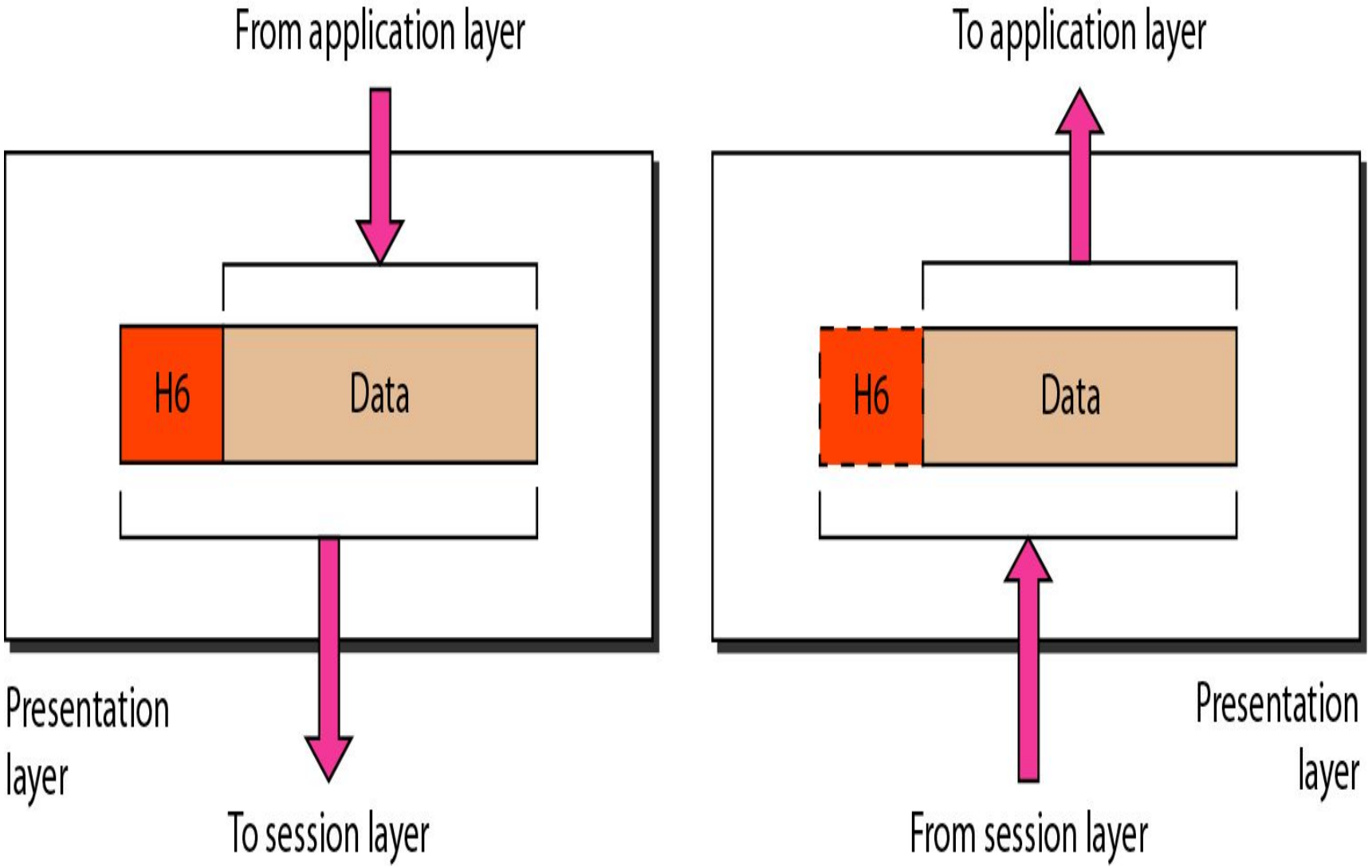
Application Layer



Presentation Layer

- The presentation layer is concerned with the syntax and semantics of the information exchanged between two systems.
- The presentation layer is responsible for;
  - **Translation;**
    - At the sender end it changes the information from its sender-dependent format into a common format.
    - At the receiving machine it changes the common format into its receiver-dependent format.
  - **Encryption; (for privacy)**
    - The sender transforms the original information to another form and sends the resulting message out over the network.
    - Decryption reverses the original process to transform the message back to its original form.
  - **Compression;**
    - Data compression reduces the number of bits contained in the information.
    - Important in the transmission of multimedia such as text, audio, and video.

Presentation Layer



Session Layer

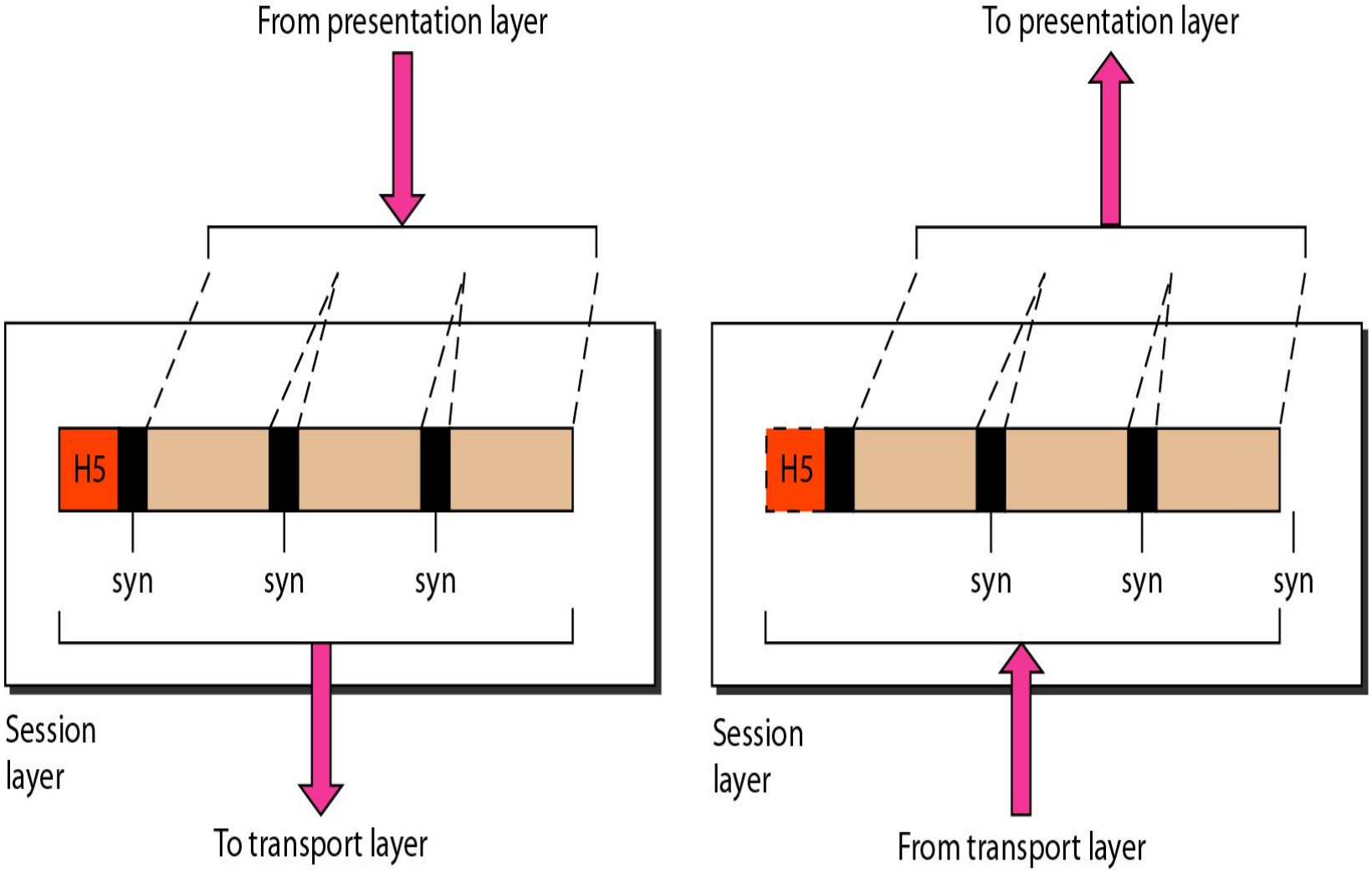
- The session layer is responsible for dialog control and synchronization.

- Responsible for;
  1. Session Creation,
  2. Session Maintenance and
  3. Synchronization.

Session Layer

- Allows two applications on different computers to establish, use and end a connection called a session.
- Provides synchronization between user tasks by placing checkpoints in the data stream.
- Also implements dialog control between communicating processes, regulating which side transmits, when, for how long, and so on.

Session Layer



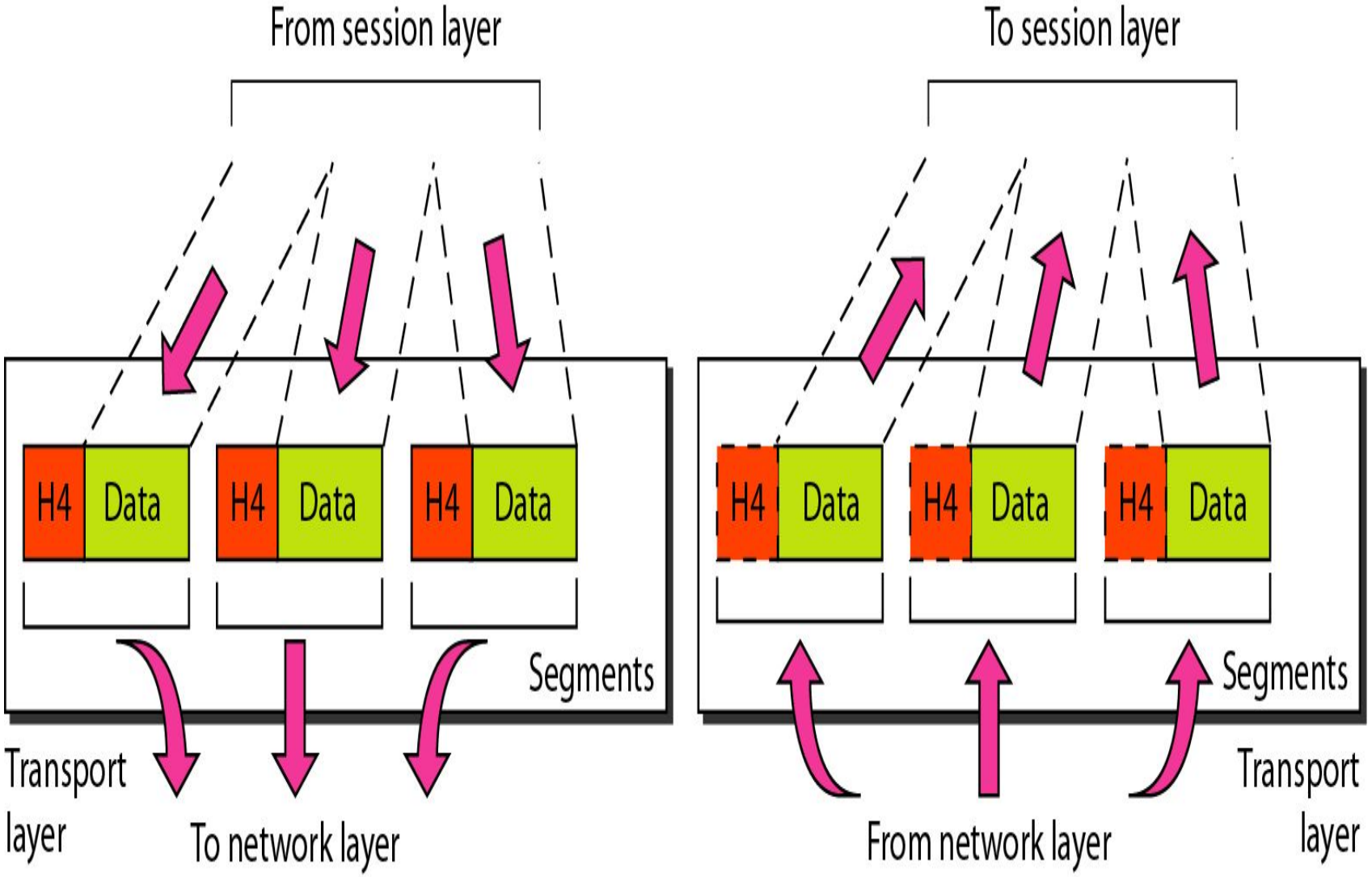
Transport Layer

- The transport layer is responsible for the delivery of a message from one process (application) to another.
- This layer ensures that packets are delivered error free, in sequence, and with no losses or duplications.
- This layer repackages messages, dividing long messages into several packets to be transmitted efficiently over the network.
- At the receiving end, the transport layer unpacks the messages, resembles the original messages and typically sends an acknowledgement of receipt.

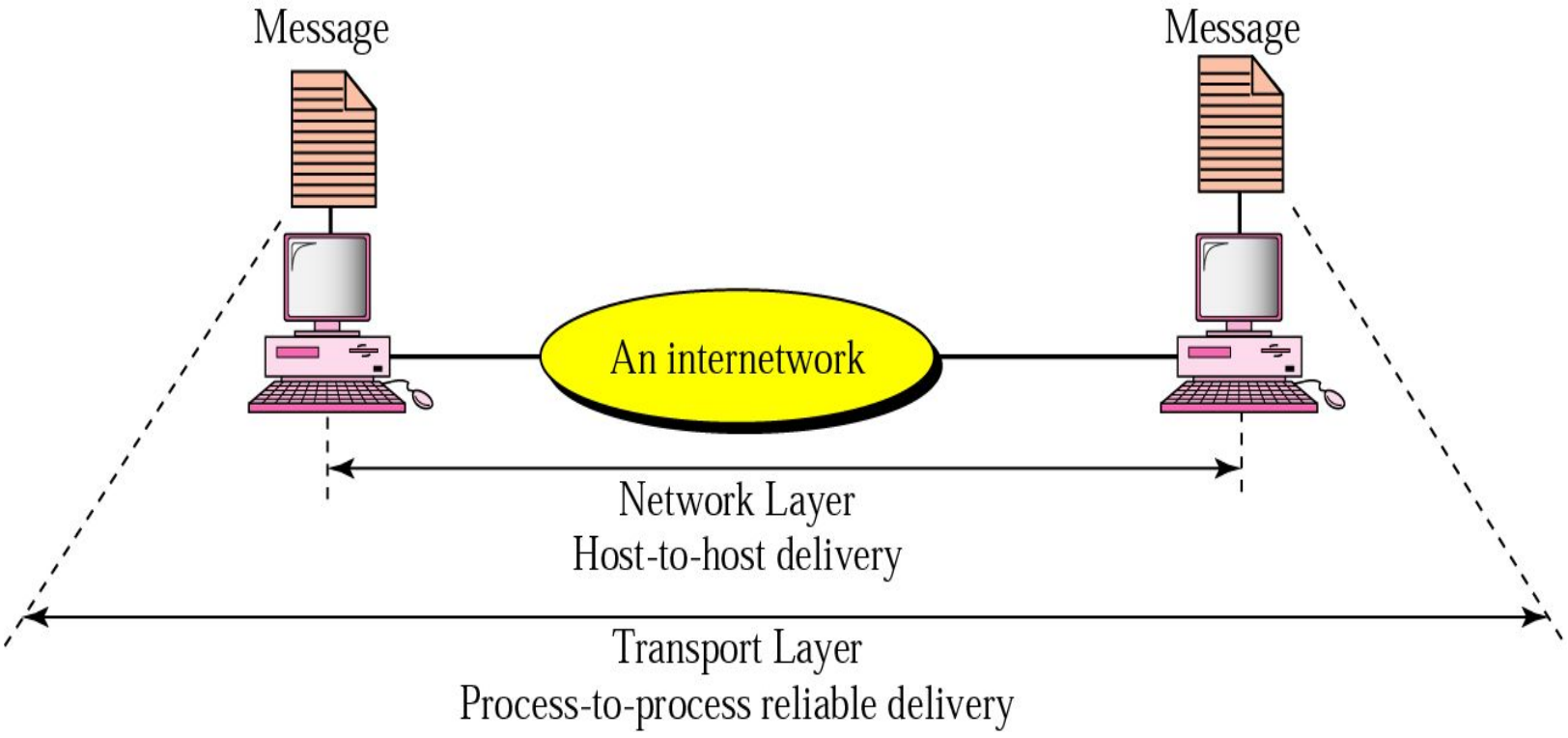
Transport Layer

- Service port addressing
- Segmentation and reassembly
- Connection control
- Flow control
- Error control

Transport Layer



Transport Layer

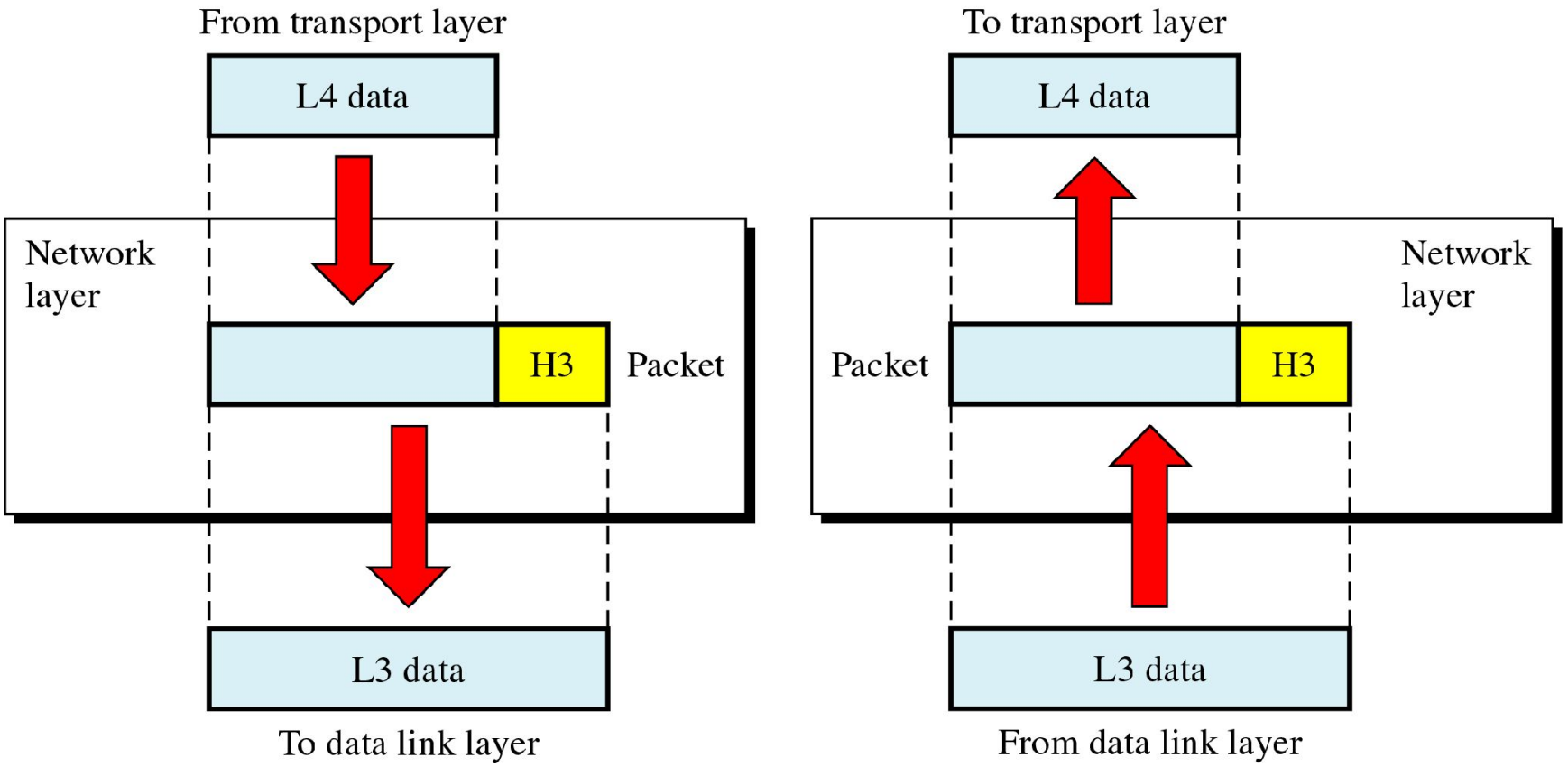


Network Layer

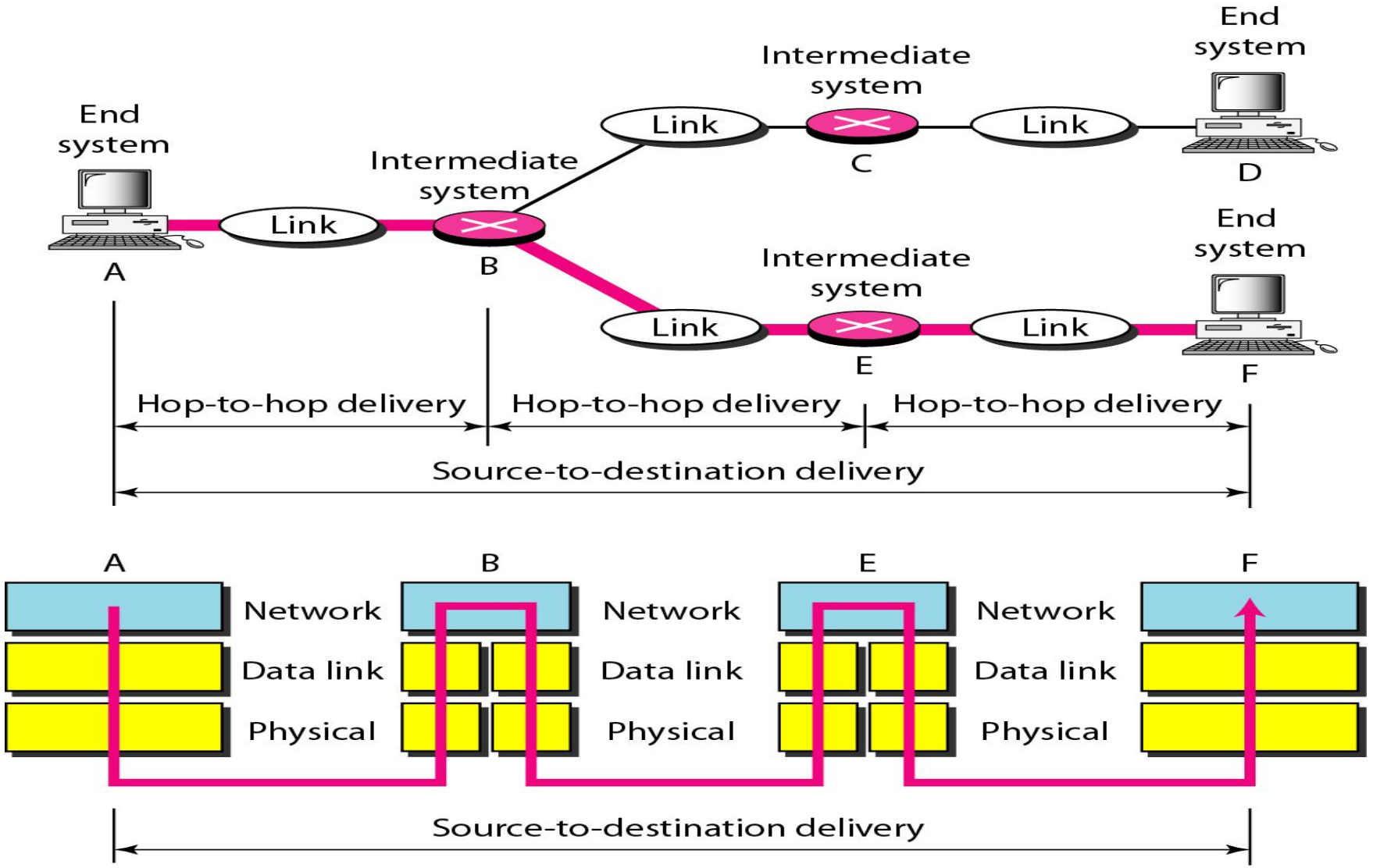
- Responsible for the delivery of individual packets from the source host to the destination host possibly across multiple networks.
- Whereas the data link layer oversees the delivery of the packet between two systems on the same network

- Responsible for;
  - Logical addressing and translating it to physical addresses.
  - Routing
    - Route Determination
    - Forwarding (packet Switching)
  - Packetizing

Network Layer



Network Layer



Data Link Layer

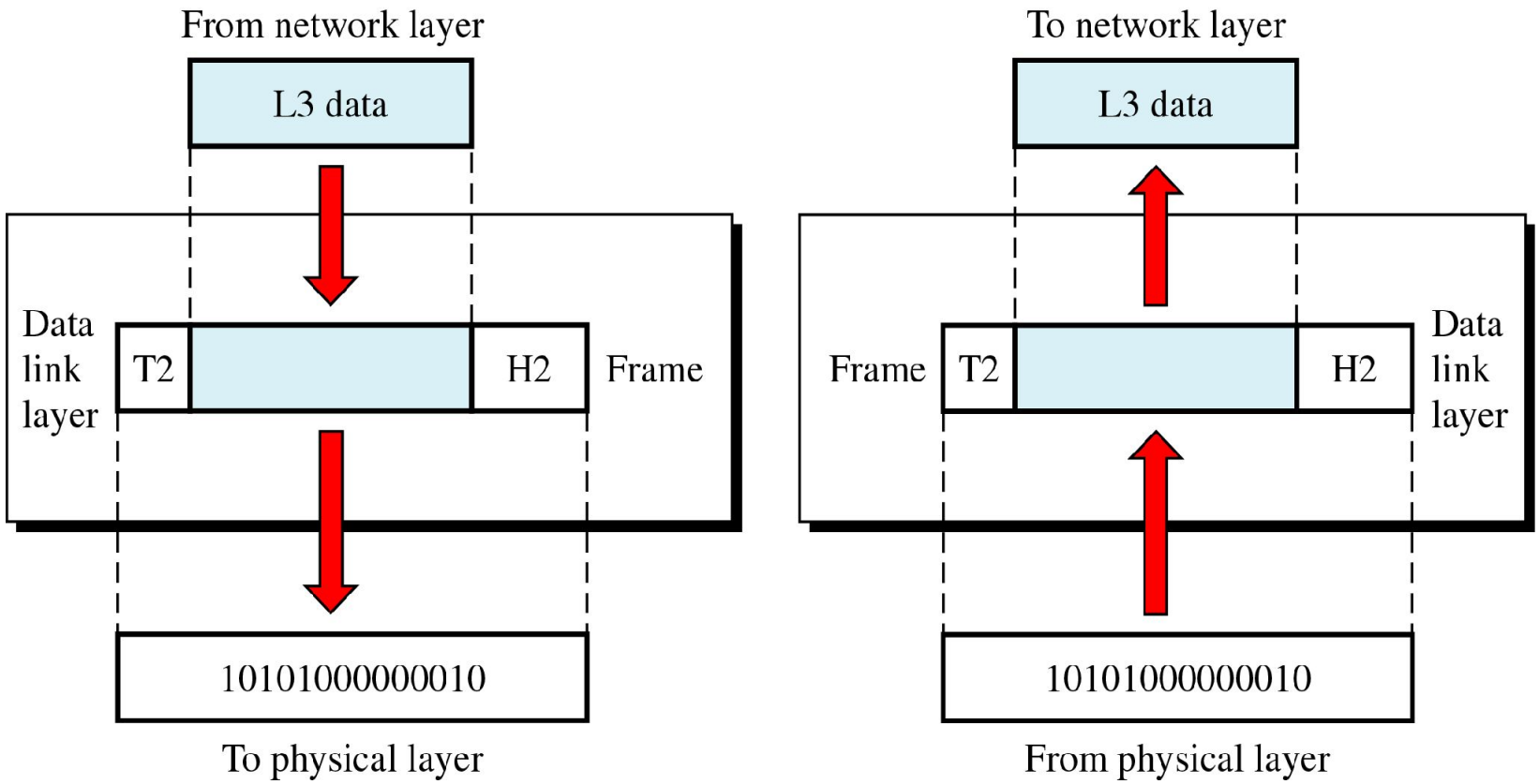
- The data link layer is responsible for moving frames from one hop (node) to the next (immediate).

- On the receiving end, it packages raw bits from the physical layer into the data frames.

Data Link Layer

- Major duties;
  - Responsible for Node-To-Node Delivery
  - Framing
  - Physical addressing (MAC)
  - Flow control (Stop & Wait, Sliding window)
  - Error control (Stop & Wait ARQ, Sliding window ARQ)
  - Access control (ENQ/ACK, POLL/SELECT)

Data Link Layer



Data Link Layer

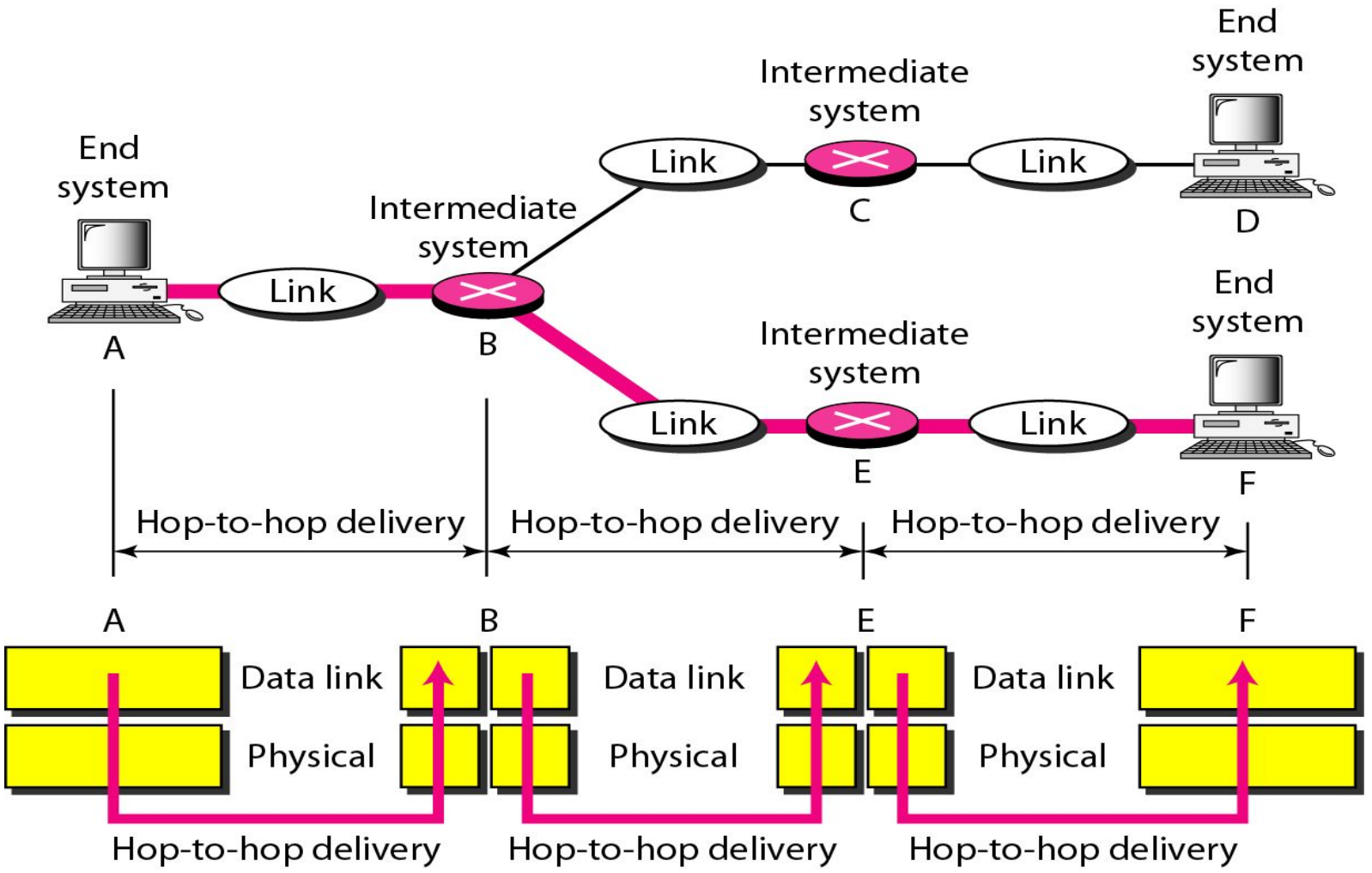
- Data Link Layer is sub-divided into two sub-layers, i.e.

1. Logical Link Control (LLC);
   - Responsible for establishing and maintaining links (i.e. either logical or physical) between the communicating devices.

2. Media Access Control (MAC).
   - Controls the way in which multiple devices share the same media channel.

Data Link Layer

- Hop-to-Hop (node-to-node) delivery



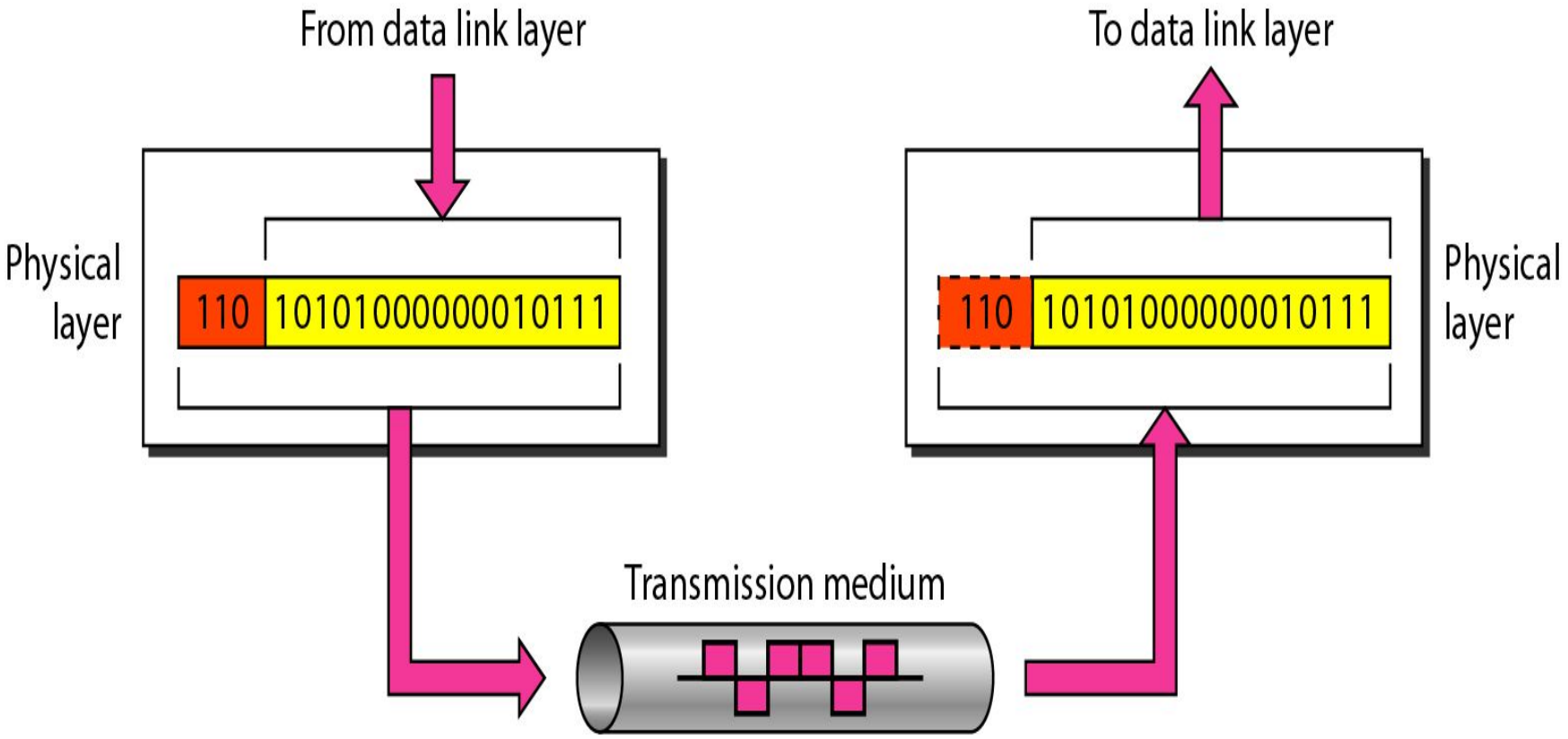
Physical Layer

- Physical layer coordinates the functions required to transmit a bit stream over a physical medium.
- It deals with the mechanical and electrical specifications of the interface and transmission medium.
- It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to occur.

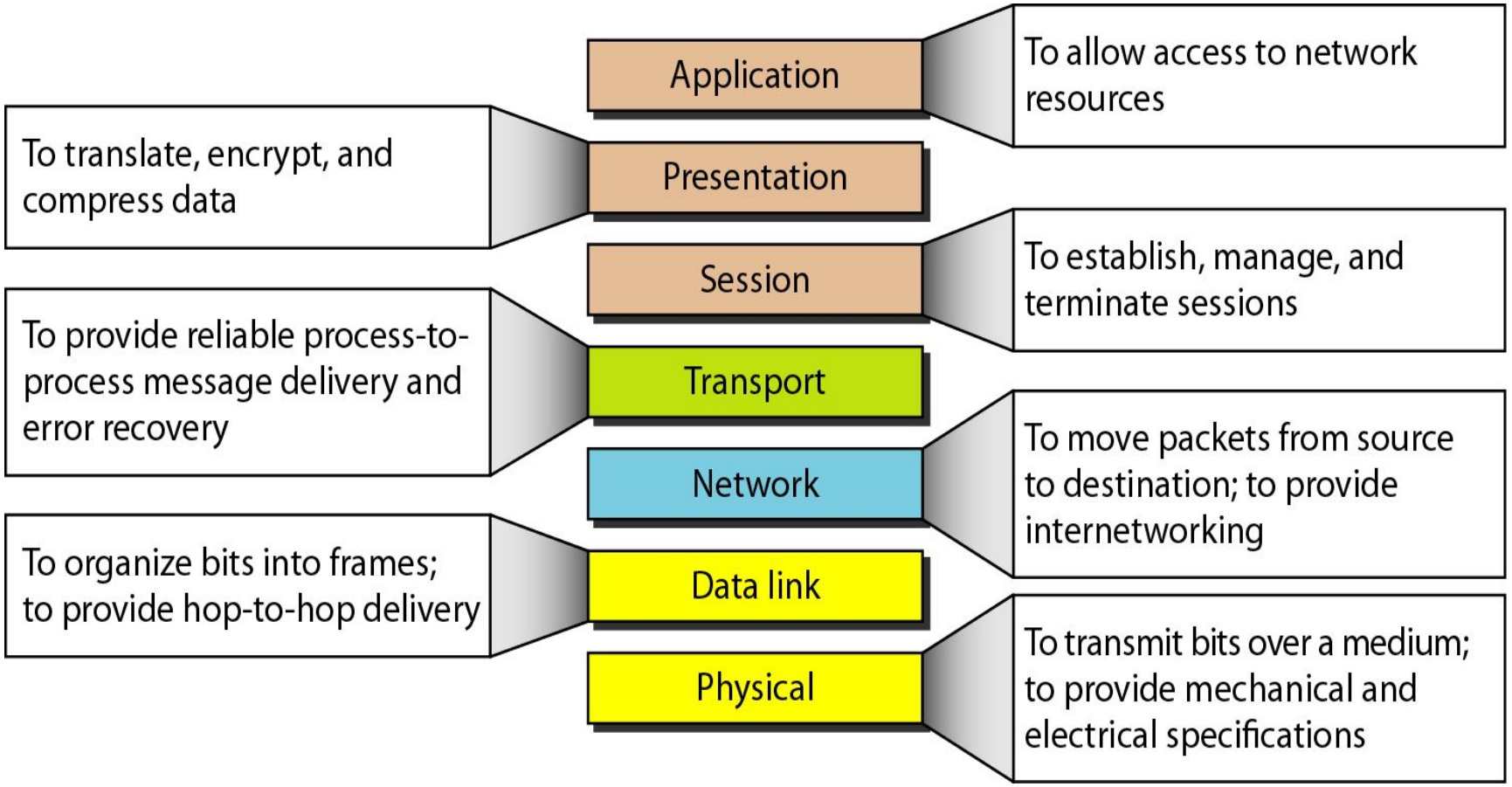
Physical Layer

- Physical layer is concerned with the following:
  - Physical characteristics of interfaces and medium
  - Representation of bits / Encoding
  - Data rate : transmission rate
  - Synchronization of bits
  - Line configuration
  - Physical topology
  - Transmission mode
  - baseband and broadband transmissions
  - Multiplexing /De multiplexing

Physical Layer



Summary of Layers



Suggested Reading

- Section
  - 2.1,
  - 2.2,
  - 2.3,
    - "Data Communications and Networking" 4th Edition by Behrouz A. Forouzan